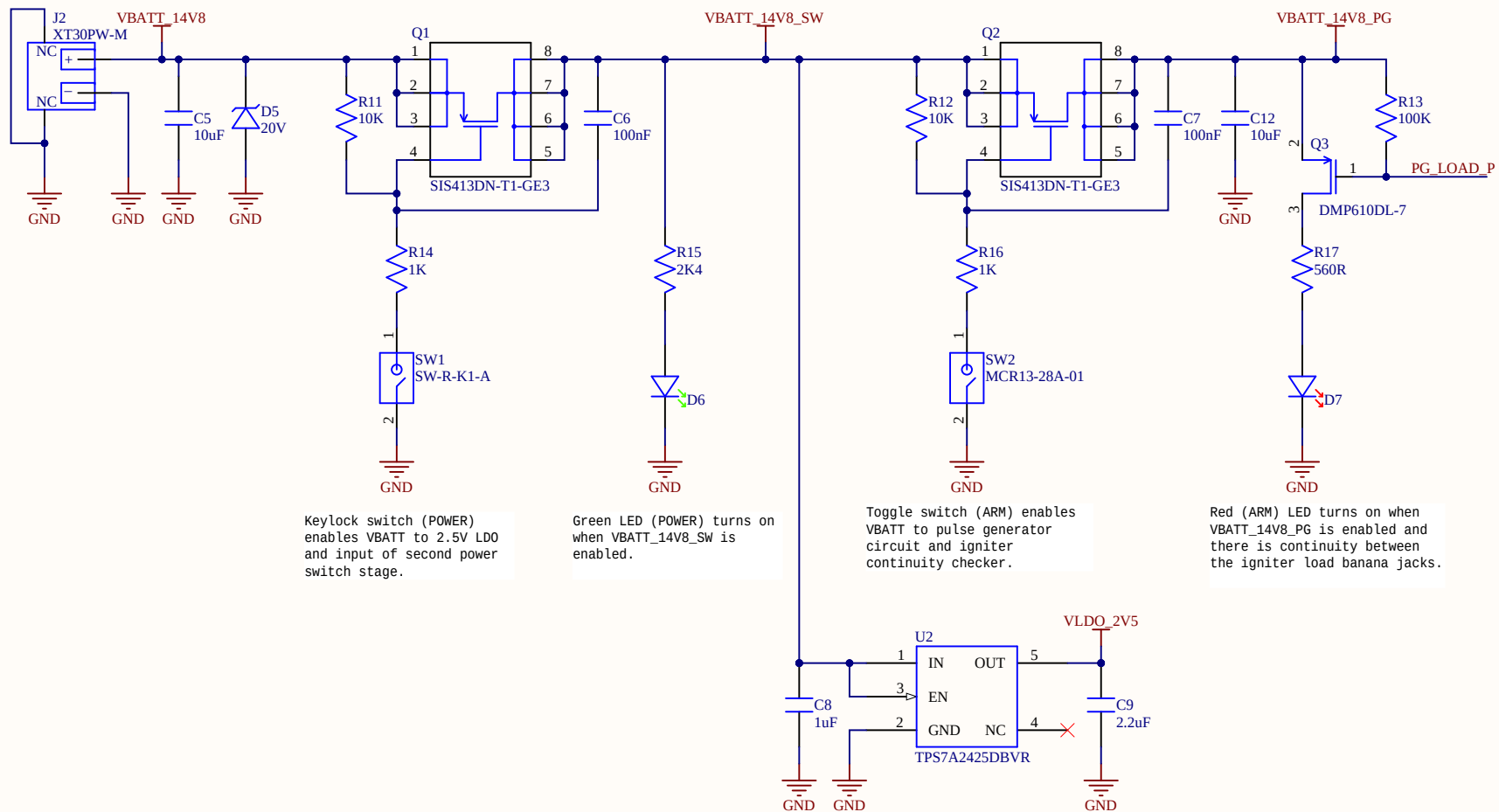




Keylock switch (POWER) enables VBATT to 2.5V LDO and input of second power switch stage.

Green LED (POWER) turns on when VBATT_14V8_SW is enabled.

Toggle switch (ARM) enables VBATT to pulse generator circuit and igniter continuity checker.

Red (ARM) LED turns on when VBATT_14V8_PG is enabled and there is continuity between the igniter load banana jacks.

LDO is used for generation of 2.5V reference voltage to pulse generator current source circuit and the LAUNCH button circuit. It powers up before the pulse generator to prevent accidental misfires during pulse generator circuit startup.

Version 1.1
Change
R15: 560R/0805->2.4K/0603 (was too bright)
R17: 680R->560R (was too dim)
C6,C7: 10nF->100nF (not debounced well enough)

Title		
Size A	Number	Revision
Date:	11-26-2024	Sheet of
File:	D:\iCloudDrive\...\Power.SchDoc	Drawn By:

A

A

B

B

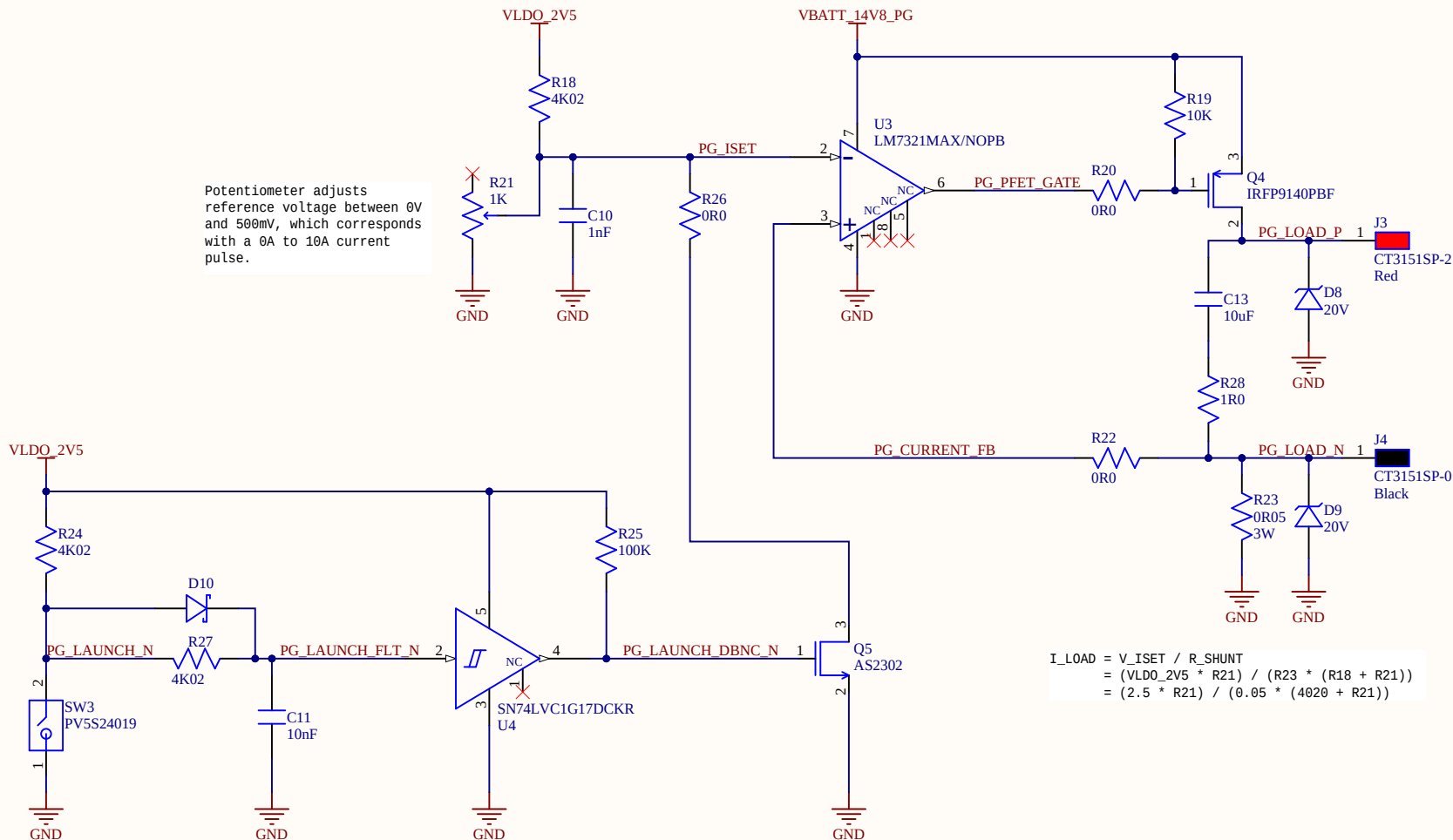
C

C

D

D

Potentiometer adjusts reference voltage between 0V and 500mV, which corresponds with a 0A to 10A current pulse.



Momentary pushbutton switch releases ISET to generate current pulse. Switch signal is debounced and buffered to drive NFET.

$$I_{LOAD} = V_{ISET} / R_{SHUNT}$$

$$= (VLD0_2V5 * R21) / (R23 * (R18 + R21))$$

$$= (2.5 * R21) / (0.05 * (4020 + R21))$$

Version
1.1

Change
C11: 3.3nF->10nF (not debounced well enough)
R20: 10R->0R0 (minor improvement to stability)
Add C13/R28 network to improve stability with inductive loads, like long wires

Title		
Size	Number	Revision
A		
Date:	11-26-2024	Sheet of
File:	D:\iCloudDrive\...\Pulse_Generator.SchDocDrawn By:	

